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Skin-attachable sensor system and method for tracking eye movement and position

Abstract

A system for tracking an eye of a user includes one or more piezoelectric sensors positionable on the face of the user, and an eye tracking computer system in signal communication with the one or more piezoelectric sensors so as to receive signals from the one or more piezoelectric sensors, wherein the computer system is configured to detect movement in at least one direction of the eye of the user based on the signals received by the computer system from the one or more piezoelectric sensors when the one or more piezoelectric sensors are positioned on the face of the user in one or more locations each spaced from the eyelids of the user.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a 35 U.S.C. § 371 national stage application of PCT/US2022/034279 filed Jun. 21 2022, entitled “Skin-Attachable Sensor System and Method for Tracking Eye Movement and Position”, which claims benefit of U.S. provisional patent application Ser. No. 63/213,048 filed Jun. 21, 2021, and entitled “Skin-Attachable Sensor System and Method For Tracking Eye Movement and Position,” which is hereby incorporated herein by reference in its entirety for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not applicable.

BACKGROUND

(2) Tracking eye movement and position may be utilized in a variety of diverse applications

including, for example, personal health and safety, public safety, medical diagnosis, and personal entertainment. With respect to personal health and safety, eye fatigue, which is commonly experienced when reading, writing, and driving for an extended period of time, has been identified as an important ocular health and safety problem. This problem is exasperated these days in view of the ever-increasing use of personal electronic devices. With respect to public safety, drowsiness and fatigue caused by, among other things, sleep deprivation, tiredness, circadian rhythm effect, and temporary brain circulation problems is a serious public safety hazard related to motor-vehicle and occupational accidents. Concerning medical diagnosis, the condition of a patient's eyes can reflect a corresponding condition of the patient's brain given that our eyes are one of the most connected organs to the brain through the body's central nervous system. Thus, the tracking of eye movement and position provides in at least some applications a simple yet efficient way to diagnose brain-related diseases in their early stages. Finally, concerning personal entertainment, eye-motion tracking has become an important component of virtual reality (VR) and augmented reality (AR) systems.

BRIEF SUMMARY OF THE DISCLOSURE

(3) An embodiment of a system for tracking an eye of a user comprises one or more piezoelectric sensors positionable on the face of the user, and an eye tracking computer system in signal communication with the one or more piezoelectric sensors so as to receive signals from the one or more piezoelectric sensors, wherein the computer system is configured to detect movement in at least one direction of the eye of the user based on the signals received by the computer system from the one or more piezoelectric sensors when the one or more piezoelectric sensors are positioned on the face of the user in one or more locations each spaced from the eyelids of the user. In some embodiments, the computer system is configured to detect the movement in the at least one direction of the eye of the user when the one or more piezoelectric sensors are positioned on the temple of the user. In some embodiments, the one or more piezoelectric sensors comprises a plurality of the piezoelectric sensors positionable on the face of the user, and the computer system is configured to detect movement of the eye of the user in both a lateral direction and a vertical direction orthogonal to the lateral direction when the plurality of the piezoelectric sensors are positioned on the face of the user in a plurality of locations each spaced from the eyelids of the user. In certain embodiments, the computer system is configured to compare the signals produced by the plurality of the piezoelectric sensors to detect the movement of the eye of the user in both the lateral direction and the vertical direction when the plurality of the piezoelectric sensors are positioned on the face of the user in the plurality of locations each spaced from the eyelids of the user. In certain embodiments, the computer system is configured to detect movement of the eye of the user in a diagonal direction that is at a non-zero angle to both the lateral direction and the vertical direction when the plurality of the piezoelectric sensors are positioned on the face of the user in the plurality of locations each spaced from the eyelids of the user. In some embodiments, the computer system is configured to detect movement of the eye of the user in a rotational direction when the plurality of the piezoelectric sensors are positioned on the face of the user in the plurality of locations each spaced from the eyelids of the user. In some embodiments, the signals produced by the one or more piezoelectric sensors are contingent upon the deflection of the facial skin of the user upon which the one or more piezoelectric sensors are positioned. In certain embodiments, each of the one or more piezoelectric sensors comprises a pair of electrodes and a piezoelectric film positioned between the pair of electrodes. In certain embodiments, the piezoelectric film comprises at least one of aluminum nitride, gallium nitride, and indium nitride. In some embodiments, each of the one or more piezoelectric sensors comprises an outer insulating layer sealing the piezoelectric film and the pair of electrodes from the external environment. In some embodiments, each of the one or more piezoelectric sensors comprises an adhesive pad for removably attaching the one or more piezoelectric sensors to the face of the user. In certain embodiments, the computer system is configured to generate an image based on the detection of the

movement of the eye of the user in the at least one direction, and the computer system comprises a visual display configured to indicate the image to the user.

(4) An embodiment of a method for tracking an eye of a user comprises (a) producing signals from one or more piezoelectric sensors in response to the user moving their eye in at least one direction, wherein the one or more piezoelectric sensors are positioned on the face of the user in one or more locations each spaced from the eyelids of the user in response to the user, (b) receiving by a computer system the signals produced by the one or more piezoelectric sensors, and (c) detecting by the computer system movement in the at least one direction of the eye of the user based on the signals received from the one or more piezoelectric sensors. In some embodiments, (a) comprises producing signals from a plurality of the piezoelectric sensors in response to the user moving their eye in at least one direction, wherein the one or more piezoelectric sensors are positioned on the face of the user in one or more locations each spaced from the eyelids of the user in response to the user, (b) comprises receiving by the computer system the signals produced by the plurality of the piezoelectric sensors, and (c) comprises detecting by the computer system movement of the eye of the user in both a lateral direction and a vertical direction orthogonal to the lateral direction. In some embodiments, (c) comprises detecting by the computer system movement of the eye of the user in a diagonal direction that is at a non-zero angle to both the lateral direction and the vertical direction. In certain embodiments, (a) comprises producing signals from a plurality of the piezoelectric sensors in response to the user moving their eye in at least one direction, wherein the one or more piezoelectric sensors are positioned on the face of the user in one or more locations each spaced from the eyelids of the user in response to the user, (b) comprises receiving by the computer system the signals produced by the plurality of the piezoelectric sensors, and (c) comprises detecting by the computer system movement of the eye of the user in a rotational direction.

(5) An embodiment of a computer system for tracking an eye of a user comprises a processor, and a storage device coupled to the processor and containing instructions that when executed cause the processor to detect movement in at least one direction of the eye of the user based on signals received by the computer system from one or more piezoelectric sensors when the one or more piezoelectric sensors are positioned on the face of the user in one or more locations each spaced from the eyelid of the user. In some embodiments, the instructions when executed cause the processor to detect movement of the eye of the user in both a lateral direction and a vertical direction orthogonal to the lateral direction when a plurality of the piezoelectric sensors are positioned on the face of the user in a plurality of locations each spaced from the eyelid of the user. In some embodiments, the instructions when executed cause the processor to detect movement of the eye of the user in a diagonal direction that is at a non-zero angle to both the lateral direction and the vertical direction when the plurality of the piezoelectric sensors are positioned on the face of the user in the plurality of locations each spaced from the eyelids of the user. In certain embodiments, the instructions when executed cause the processor to detect movement of the eye of the user in a rotational direction when a plurality of the piezoelectric sensors are positioned on the face of the user in a plurality of locations each spaced from the eyelids of the user.

(6) An embodiment of a flexible piezoelectric sensor comprises a pair of electrically conductive electrodes, a piezoelectric film positioned between the pair of electrodes, the piezoelectric film comprising a Gallium Nitride material having a single-crystalline structure, and an electrical insulator sealing the pair of electrodes and the piezoelectric film from the surrounding environment, wherein the pair of electrodes are configured to produce an output voltage in response to a deflection of the piezoelectric film. In some embodiments, the Gallium Nitride material comprises a III-nitride material. In some embodiments, the piezoelectric film comprises at least one of Gallium Nitride (GaN), Aluminum Nitride (AlN), Scandium Nitride (ScN), and Indium Nitride (InN) in accordance with the formula of $\text{In}_x\text{Al}_y\text{Sc}_z\text{Ga}_{1-x-y}\text{N}$, where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$. In some embodiments, the thickness of the piezoelectric film is between 1.0 nanometer

(nm) and 1.0 millimeter (mm). In certain embodiments, the piezoelectric sensor has a sensitivity between 0.5 volts per newton (V/N) and 5 V/N. In certain embodiments, the Gallium Nitride material has an outer sidewall having a surface roughness of 500 micrometers (μm) or less. In certain embodiments, the piezoelectric film is lead free. In some embodiments, the piezoelectric sensor comprises an adhesive pad coupled to the electrical insulator for releasably attaching the piezoelectric sensor to a surface.

(7) An embodiment of a flexible piezoelectric sensor comprises a pair of electrically conductive electrodes, a piezoelectric film positioned between the pair of electrodes, the piezoelectric film comprising a Gallium material, and an electrical insulator sealing the pair of electrodes and the piezoelectric film from the surrounding environment, wherein the pair of electrodes are configured to produce an output voltage in response to a deflection of the piezoelectric film, wherein the piezoelectric sensor has a sensitivity between 0.1 volts per newton (V/N) and 5 V/N. In some embodiments, the sensitivity of the piezoelectric sensor is between 0.1 V/N and 1 V/N. In some embodiments, the Gallium Nitride material has an outer sidewall having a surface roughness of 500 micrometers (μm) or less. In certain embodiments, the surface roughness of the Gallium Nitride material is 100 μm or less. In certain embodiments, the Gallium Nitride material has a single-crystalline structure. In some embodiments, the piezoelectric film comprises at least one of Gallium Nitride (GaN), Aluminum Nitride (AlN), Scandium Nitride (ScN), and Indium Nitride (InN).

(8) An embodiment of a method for forming a flexible piezoelectric sensor comprises (a) growing a piezoelectric film on a first substrate, wherein the piezoelectric film comprises a Gallium material and has an outer edge extending along the perimeter of the film, (b) trimming at least a portion of the outer edge from the piezoelectric film, and (c) coupling a pair of electrically conductive electrodes to the trimmed piezoelectric film whereby the trimmed piezoelectric film is positioned between the pair of electrodes. In some embodiments, the piezoelectric film comprises at least one of Gallium Nitride (GaN), Aluminum Nitride (AlN), Scandium Nitride (ScN) and Indium Nitride (InN). In some embodiments, the outer edge trimmed from the piezoelectric film is at least 0.5 millimeters (mm) in width. In certain embodiments, the method comprises (d) attaching a second substrate to one of the pair of electrodes, and (e) removing the first substrate from the piezoelectric film. In certain embodiments, the first substrate comprises Silicon and the second substrate comprises Sapphire. In some embodiments, the method comprises (d) enclosing the pair of electrodes and the trimmed piezoelectric film with an electrical insulator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a detailed description of exemplary embodiments of the disclosure, reference will now be made to the accompanying drawings in which:

(2) FIG. 1 is a schematic view of an embodiment of an eye tracking system;

(3) FIG. 2 is a perspective view of an embodiment of a piezoelectric sensor of the eye tracking system of FIG. 1;

(4) FIG. 3 is a graph illustrating an output voltage of a piezoelectric sensor as a function of time;

(5) FIGS. 4 and 5 are schematic views of an output voltage response of a plurality of piezoelectric sensors responding to movement of a user's eye;

(6) FIG. 6 is a graph illustrating an output voltage of a first piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 4 as a function of time;

(7) FIG. 7 is a graph illustrating an output voltage of a second piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 4 as a function of time;

(8) FIG. 8 is a graph illustrating an output voltage of a third piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 4 as a function of time;

(9) FIG. 9 is a graph illustrating an output voltage of a first piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 5 as a function of time;

(10) FIG. 10 is a graph illustrating an output voltage of a second piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 5 as a function of time;

(11) FIG. 11 is a graph illustrating an output voltage of a third piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 5 as a function of time;

(12) FIGS. 12 and 13 are schematic views of an output voltage response of a plurality of piezoelectric sensors responding to movement of a user's eye;

(13) FIG. 14 is a graph illustrating an output voltage of a first piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 12 as a function of time;

(14) FIG. 15 is a graph illustrating an output voltage of a second piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 12 as a function of time;

(15) FIG. 16 is a graph illustrating an output voltage of a third piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 12 as a function of time;

(16) FIG. 17 is a graph illustrating an output voltage of a first piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 13 as a function of time;

(17) FIG. 18 is a graph illustrating an output voltage of a second piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 13 as a function of time;

(18) FIG. 19 is a graph illustrating an output voltage of a third piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 13 as a function of time;

(19) FIG. 20 is a schematic view of an output voltage response of a plurality of piezoelectric sensors responding to movement of a user's eye;

(20) FIG. 21 is a graph illustrating an output voltage of a first piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 20 as a function of time;

(21) FIG. 22 is a graph illustrating an output voltage of a second piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 20 as a function of time;

(22) FIG. 23 is a graph illustrating an output voltage of a third piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 20 as a function of time;

(23) FIG. 24 is a schematic view of an output voltage response of a plurality of piezoelectric sensors responding to movement of a user's eye;

(24) FIG. 25 is a graph illustrating an output voltage of a first piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 24 as a function of time;

(25) FIG. 26 is a graph illustrating an output voltage of a second piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 24 as a function of time;

(26) FIG. 27 is a graph illustrating an output voltage of a third piezoelectric sensor of the plurality of piezoelectric sensors of FIG. 24 as a function of time;

(27) FIG. 28 is a schematic side view of an embodiment of a flexible piezoelectric sensor;

(28) FIGS. 29-31 are schematic side views of an exemplary process for forming a flexible piezoelectric sensor;

(29) FIG. 32 is a flowchart of an embodiment of a method for tracking an eye of a user;

(30) FIG. 33 is a flowchart of an embodiment of a method for forming a flexible piezoelectric sensor;

(31) FIG. 34 is a schematic view of an exemplary process for forming a flexible piezoelectric sensor;

(32) FIG. 35 is a schematic view of an exemplary flexible piezoelectric sensor;

(33) FIG. 36 is a graph illustrating X-ray diffraction characterization of a single-crystalline III-Nitride thin film along the GaN (0002) plane;

(34) FIG. 37 is a graph illustrating X-ray diffraction characterization of a single-crystalline III-Nitride thin film along the GaN (10-12) plane;

(35) FIG. 38 is a side cross-sectional view of an exemplary flexible piezoelectric sensor;

(36) FIG. 39 is a schematic of a flexible piezoelectric sensor in different states;
(37) FIG. 40 is a graph illustrating a calculated output voltage from a flexible piezoelectric sensor as a function of simulated skin deflection;
(38) FIG. 41 is a perspective view of an embodiment of a piezoelectric sensor;
(39) FIG. 42 is a schematic of different regions of a user's face; and
(40) FIGS. 43-55 illustrate output voltages produced by exemplary flexible piezoelectric sensors positioned on a user's face.

DETAILED DESCRIPTION

(41) The following discussion is directed to various exemplary embodiments. However, one skilled in the art will understand that the examples disclosed herein have broad application, and that the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to suggest that the scope of the disclosure, including the claims, is limited to that embodiment.

(42) Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not function. The drawing figures are not necessarily to scale. Certain features and components herein may be shown exaggerated in scale or in somewhat schematic form and some details of conventional elements may not be shown in interest of clarity and conciseness.

(43) In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . .” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection, or through an indirect connection via other devices, components, and connections. In addition, as used herein, the terms “axial” and “axially” generally mean along or parallel to a central axis (e.g., central axis of a body or a port), while the terms “radial” and “radially” generally mean perpendicular to the central axis. For instance, an axial distance refers to a distance measured along or parallel to the central axis, and a radial distance means a distance measured perpendicular to the central axis. As used herein, the terms “approximately,” “about,” “substantially,” and the like mean within 10% (i.e., plus or minus 10%) of the recited value. Thus, for example, a recited angle of “about 80 degrees” refers to an angle ranging from 72 degrees to 88 degrees.

(44) As described above, tracking of eye movement and position provides advantages in a wide variety of applications including, among others, personal health and safety, public safety, medical diagnosis, and personal entertainment. Conventional systems for tracking eye movement and position typically rely on computer vision-based approaches which can be broken down into remote eye tracking systems and head-mount-display (HMD)-embedded eye tracking systems. Remote eye tracking systems typically require expensive and bulky components such as a high-resolution camera, mounting units, and associated image processing software, generally preventing such systems from achieving miniaturization. Additionally, remote eye tracking systems require the localization of the user's head and eyes which may not be practical in applications in which the user's head moves relative to the camera of the remote eye tracking system. HMD-embedded eye tracking systems may address some of these challenges but introduce additional challenges in the form of cumbersome, uncomfortable, and expensive headgear worn by the user. Moreover, HMD-embedded eye tracking systems may not be employable in some applications in which the user cannot wear the eye tracking headgear such as when operating a motor vehicle.

(45) Outside of computer vision-based approaches, another technique for tracking eye movement and position is electrooculography (EOG) in which a potential difference (the “EOG signal”) between the cornea and retina of the user's eye is monitored and correlated with the position of the user's eye. However, the EOG signal relied upon for determining eye position is generally weak and thus easily influenced by external conditions. Thus, EOG based approaches are typically

limited to applications under which the eye tracking is performed under controlled conditions such as with some medical diagnoses.

(46) Accordingly, embodiments disclosed herein include systems and methods for tracking a user's eye using one or more piezoelectric sensors positionable on the user's face at one or more locations that are spaced from the user's eyelids and thus may be comfortably worn by the user. Additionally, embodiments of flexible piezoelectric sensors and methods for forming flexible piezoelectric sensors are described herein. The piezoelectric sensors may be formed from nontoxic (lead free) materials and thus may be safely worn by the user without posing risk to the user's health. The piezoelectric sensors have a sensitivity sufficiently great enough to detect the movement of the facial skin upon which the piezoelectric sensors are positioned such that the sensors produce signals (e.g., one or more output voltages) that varies in response to movement of the facial skin. The piezoelectric sensor may comprise a Gallium Nitride material having a single-crystalline structure providing the piezoelectric sensor with both sufficient flexibility and sensitivity. As will be described further herein, the Gallium Nitride material may comprise a III-Nitride material including Aluminum Nitride and Aluminum Gallium Nitride. Additionally, a piezoelectric thin film of the sensor comprising the Gallium Nitride material may be trimmed during the process of forming the sensor to free the thin film from microcracks or other flaws which tend to occur along the outer edge of the thin film, thereby maximizing the durability of the piezoelectric sensor.

(47) Embodiments of eye tracking systems disclosed herein include an eye tracking computer system which receives signals from the one or more piezoelectric sensors and detects movement of the user's eye in one or more directions based on the received signals. As will be described further herein, eye tracking systems disclosed herein may include a plurality of piezoelectric sensors positioned at different locations on the user's face (each spaced from the eyelids of the user) to permit the system to detect movement both laterally, vertically, diagonally, and rotationally. This information (e.g., current direction and/or rate of travel of the eye, current position of the eye, etc.) may be utilized in generating one or more images by the eye tracking system which may then be indicated to the user by the eye tracking computer system. For example, the position and/or movement of the user's eye may be utilized in generating a VR or AR environment.

(48) Referring now to FIG. 1, an embodiment of an eye tracking system **10** for tracking the movement and position of an eye **2** of a user **1** is shown. In this exemplary embodiment, eye tracking system **10** generally includes an eye tracking computer system **20** and a plurality of flexible, skin-attachable, flexible piezoelectric sensors **100A-100C** which form a sensor network **101** of the eye tracking system **10**. As will be described further herein, computer system **20** tracks the movement and position of the user's eye **2** based on signals provided to the computer system **20** from the plurality of piezoelectric sensors **100A-100C**. Computer system **20** is thus in signal communication with the plurality of piezoelectric sensors **100A-100C** via a signal connection or pathway **15** formed therebetween. The signal connection **15** may be wired with computer system **20** located proximal the plurality of piezoelectric sensors **100A-100C**. Alternatively, signal connection **15** may be wireless with computer system **20** located proximal or distal (or both with a portion of system **20** proximal and a portion of system **20** distal) the plurality of piezoelectric sensors **100A-100C**. For example, the plurality of piezoelectric sensors **100A-100C** may each include a wireless transmitter for communicating wirelessly with the computer system **20**.

(49) In this exemplary embodiment, computer system **20** generally includes a processor **22** (which may be referred to as a central processor unit or CPU) that is in communication with one or more memory devices **26**, and input/output (I/O) devices **30**. The processor **22** may be implemented as one or more CPU chips. The memory devices **26** of computer system **20** may include secondary storage (e.g., one or more disk drives, etc.), a non-volatile memory device such as read only memory (ROM), and a volatile memory device such as random-access memory (RAM). In some contexts, the secondary storage ROM, and/or RAM comprising the memory devices **26** of computer system **20** may be referred to as a non-transitory computer readable medium or a

computer readable storage media. I/O devices **30** may include printers, video monitors, liquid crystal displays (LCDs), touch screens, keyboards, keypads, switches, dials, mice, and/or other well-known input devices. Although shown as including a single CPU **22**, and a single memory device **26**, it may be understood that computer system **20** may include a plurality of separate CPUs **22**, memory devices **26**, and I/O devices **30**. It may also be understood that computer system **20** may be embodied in a networked computing system such as a cloud computing environment in which, for example, components of computer system **20** are executed and/or stored in the cloud rather than locally on a single computer.

(50) It is understood that by programming and/or loading executable instructions onto the computer system **20**, at least one of the CPU **22**, the memory devices **26** are changed, transforming the computer system **20** in part into a particular machine or apparatus having the novel functionality taught by the present disclosure. Additionally, after the computer system **20** is turned on or booted, the CPU **22** may execute a computer program or application. For example, the CPU **22** may execute software or firmware stored in the memory devices **26**. During execution, an application may load instructions into the CPU **22**, for example load some of the instructions of the application into a cache of the CPU **22**. In some contexts, an application that is executed may be said to configure the CPU **22** to do something, e.g., to configure the CPU **22** to perform the function or functions promoted by the subject application. When the CPU **22** is configured in this way by the application, the CPU **22** becomes a specific purpose computer or a specific purpose machine.

(51) Piezoelectric sensors **100A-100C** of eye tracking system **10** are positionable on a face **3** of the user **1** to monitor physiological phenomena of the user **1**. In this exemplary embodiment, piezoelectric sensors **100A-100C** comprise piezoelectric sensors which, when positioned on the user's face **3**, produce electrical signals in response to flexing or other movement of the user's facial skin in response to the user **1** moving their eye **2**. Piezoelectric sensors **100A-100C** may thus also be referred to herein as piezoelectric sensors **100A-100C**. As will be discussed further herein, piezoelectric sensors **100A-100C** each comprise a piezoelectric film including Gallium Nitride, Aluminum Gallium Nitride, and Aluminum Nitride. The piezoelectric sensors **100A-100C** may be removably or releasably attached in a comfortable and safe manner to the user's face **3** through a variety of means including, for example, adhesive pads, suction cups, or other releasable connectors.

(52) It may be understood that movement of the eye **2** is controlled by a network of several distinct muscles known as the extraocular muscles. The action of the extraocular muscles produces, along with movement of the user's eye **2**, movement in the user's skin positioned around the extraocular muscles. This movement of the user's skin may result in the mechanical deflection of a piezoelectric sensor **100** (e.g., one of piezoelectric sensors **100A-100C** shown in FIG. **1**) positioned on the moving skin, thereby causing the piezoelectric sensor **100** to produce or modulate an electrical signal. The degree of mechanical deflection produced in a given piezoelectric sensor **100** is contingent upon the direction of movement of the user's eye **2** (dependent upon which of the extraocular muscles are activated) and the position of the given piezoelectric sensor **100** on the user's face **3**. In this exemplary embodiment, a first or upper piezoelectric sensor **100A** is positioned on an upper temple area **4**, a second or middle piezoelectric sensor **100B** is positioned on a middle temple area **5**, and a third or lower piezoelectric sensor **100C** is positioned on a lower temple area **6** of the user's face **3**. However, it may be understood that piezoelectric sensors **100A-100C** may be positioned at various locations on the user's face **3** such as user's forehead, the user's cheek and other locations upon which a given piezoelectric sensor **100** of eye tracking system **10** may be safely and comfortably worn by the user **1**. In other words, sensitive parts of the user's face **3** such as the user's eyelid **7** may be avoided when positioning piezoelectric sensors **100A-100C** on the user's face **3**.

(53) Referring to FIGS. **1-3**, aspects of the operation of eye tracking system **10** are illustrated schematically in FIGS. **2** and **3**. Particularly, FIG. **2** illustrates schematically one of the

piezoelectric sensors **100** of eye tracking system **10** in an initial state **103**, a first flexed state **105**, and a third flexed state **107**. The initial state **103** of piezoelectric sensor **100** corresponds to the state or shape of piezoelectric sensor **100** when piezoelectric sensor is positioned on the temple area (e.g., one of temple areas **4**, **5**, and **6** of user **1**) of a user with the user looking centrally or straight ahead. The first flexed state **105** of piezoelectric sensor **100** corresponds to the user moving their eye laterally in the direction of the temple area (with the piezoelectric sensor **100** placed on the temple area), while the second flexed state **107** corresponds to the user moving their eye laterally in the opposing direction away from the temple area. As can be seen in FIG. 2, piezoelectric sensor **100** is deflected or contracted as it transitions from the initial state **103** to the first flexed state **105** such that a degree of curvature along the piezoelectric sensor **100** increases as the piezoelectric sensor **100** moves from the initial state **103** to the first flexed state **105**.

Conversely, piezoelectric sensor **100** is deflected as it transitions from the initial state **103** to the second flexed state **107** whereby the degree of curvature along the sensor **100** decreases as the sensor **100** moves from the initial state **103** to the second flexed state **107**.

(54) The change in deflection along piezoelectric sensor **100** alters an electrical voltage produced by the sensor **100**. Particularly, a change in deflection along sensor **100** may change both a magnitude and a polarity of the electrical voltage produced by piezoelectric sensor **100**. FIG. 3 illustrates a graph **150** of electrical voltage signal or output **152** produced by the sensor **100** as a user, beginning in an initial pose looking centrally straight ahead, alternately looks laterally to the right (towards the piezoelectric sensor **100** positioned on the user's temple) and left (away from the piezoelectric sensor **100**). As shown by graph **150**, the user moving his eye towards the right (transitioning piezoelectric sensor **100** into the first flexed state **105**) produces a series of positive polarity peaks **153** of the voltage output **152**. Conversely, the user moving his eye towards the left (transitioning sensor **100** into the second flexed state **107**) produces a series of negative polarity peaks **154** (having an opposed plurality to the positive polarity peaks **153**) of the voltage output **152**. However, the voltage output **152** of piezoelectric sensor **100** is substantially zero when sensor **100** is in the initial state **103**. Thus, further bending of piezoelectric sensor **100** from the initial state **103** (e.g., into the first flexed state **105**) results in sensor **100** producing a positive electrical voltage, while the straightening out of sensor **100** from the initial state **103** (e.g., into the second flexed state **107**) produces a negative electrical voltage from the sensor **100**. Graph **150** thus illustrates how the magnitude and polarity of the voltage produced the piezoelectric sensor **100**. By correlating the magnitude and polarity of the voltage output of the piezoelectric sensor **100** with the current position of the user's eye, the position of the user's eye may be monitored or tracked via the electrical voltage produced by sensor **100**.

(55) While FIGS. 2 and 3 illustrate that a single piezoelectric sensor **100** may be used to track the position of a user's eye in the lateral direction (moving between left and right), the eye tracking system **10** is configured to track the position of the user's eye **2** in both the lateral and vertical directions, as well as combinations thereof (e.g., in a diagonal direction by looking towards the "corner" of the eye, in a rotational direction by "rolling" the eye, etc.). This is accomplished by integrating the signals produced by two or more separate piezoelectric sensors (e.g., piezoelectric sensors **100A-100C**) attached to the user's face at different locations to form a sensor network.

(56) Referring now to FIGS. 1 and 4-11, additional aspects of the operation of eye tracking system **10** are illustrated schematically in FIGS. 4-11. Particularly, FIGS. 4 and 5 illustrate exemplary electrical voltage signals or outputs **161A-161C** for piezoelectric sensors **100A** (positioned on upper temple area **4**), **100B** (positioned on middle temple area **5**), and **100C** (positioned on lower temple area **6**), respectively, in response to a user **160** looking centrally straight ahead (indicated by arrow **162** in FIGS. 4 and 5), laterally to the left (indicated by arrow **163** in FIG. 4), laterally to the right (indicated by arrow **164** in FIG. 4), vertically upwards (indicated by arrow **165** in FIG. 5), and vertically downwards (indicated by arrow **166** in FIG. 5).

(57) In this example, when the user's eye moves, the deflection of skin behaves similarly as

described earlier with respect to FIGS. 3 and 4; however, the degree of deflection in different sensing locations (the locations at which the sensors **100A-100C** are attached) are different, resulting in the different magnitude of voltage signals **161A-161C** from each piezoelectric sensor **100A-100C**, respectively. In the case of lateral eye movement shown in FIG. 4, the muscles around the user's eye, such as the lateral rectus and medial rectus, are mainly used. The muscles and ligaments underneath the temple area of the user **160** are contracted and expanded depending on the direction of the lateral eye movements. For pure lateral movement shown in FIG. 4, the magnitude of the voltage output **161B** of middle piezoelectric sensor **100B** is greater than the voltage outputs **161A** and **161C** of piezoelectric sensors **100A** and **100C**, respectively, which are similar to each other. Graphs **170**, **175**, and **180** shown in FIGS. 6-8, respectively, illustrate voltage output **161A** (graph **170**), voltage output **161B** (graph **175**), and voltage output **161C** (graph **180**) as the user **160** looks alternately in the lateral left and right directions as indicated in FIG. 4. As shown by graphs **170**, **175**, and **180**, the magnitude of both the positive and negative polarity peaks of the voltage output **176** of middle piezoelectric sensor **100B** is greater than the positive and negative polarity peaks of voltage outputs **171** and **181**. It may be observed from these examples that the piezoelectric sensor **100** most greatly aligned with the direction of movement of the eye or in closest proximity to the position of the eye (middle piezoelectric sensor **100B** in the example of FIGS. 4 and 6-8) has the great voltage output.

(58) For a pure vertically upward movement shown in FIG. 5, the magnitude of the voltage output **161A** of upper piezoelectric sensor **100A** is greater than the voltage outputs **161B** and **161C** of piezoelectric sensors **100B** and **100C**, respectively, while the voltage output **161B** of piezoelectric sensor **100B** is greater than voltage output **161C** of piezoelectric sensor **100C**. Conversely, for a pure vertically downward movement also shown in FIG. 5, the magnitude of the voltage output **161C** of lower piezoelectric sensor **100C** is greater than the voltage outputs **161B** and **161A** of piezoelectric sensors **100B** and **100A**, respectively, while the voltage output **161B** of piezoelectric sensor **100B** is greater than voltage output **161A** of piezoelectric sensor **100A**. Graphs **185**, **190**, and **195** shown in FIGS. 9-11, respectively, illustrate voltage output **161A** (graph **185**), voltage output **161B** (graph **190**), and voltage output **161C** (graph **195**) as the user **160** looks alternately in the vertically upwards and downwards directions as indicated in FIG. 5. As shown by graph **185**, the magnitude of the positive polarity peaks (corresponding to the user **160** looking vertically upwards) of voltage output **161A** is greater than the magnitude of the negative polarity peaks **187** of voltage output **161A** (corresponding to the user **160** looking vertically downwards). Conversely, and as shown by graph **195**, the magnitude of the positive polarity peaks (corresponding to the user **160** looking vertically upwards) of voltage output **161C** is less than the magnitude of the positive polarity peaks of voltage output **161C** (corresponding to the user **160** looking vertically downwards). Finally, as shown by graph **190**, the magnitude of the positive polarity peaks of voltage output **161B** is substantially equal to the magnitude of the negative polarity peaks of voltage output **161B**.

(59) Referring now to FIGS. 1 and 12-27, additional aspects of the operation of eye tracking system **10** are illustrated schematically in FIGS. 12-27. As described above, eye tracking system **10** may track eye movements that are not either purely lateral (shown in FIG. 4) or purely vertical (as shown in FIG. 5). The tracking of eye movement can be extended to other movements including, for example, diagonal and rotational directions with distinguishable signals from the piezoelectric sensors **100A-100C** depending on their position on the face of the user **160**. FIGS. 12 and 13 illustrate exemplary electrical voltage signals or outputs **161A-161C** for piezoelectric sensors **100A** (positioned on upper temple area **4**), **100B** (positioned on middle temple area **5**), and **100C** (positioned on lower temple area **6d**) in response to the user **160** looking alternately between a first diagonal direction (indicated by arrow **200** in FIG. 12) and a second diagonal direction (indicated by arrow **201** in FIG. 12), and the user **160** looking alternately between a third diagonal direction (indicated by arrow **202** in FIG. 13) and a fourth diagonal direction (indicated by

arrow **203** in FIG. **13**). The first diagonal direction **200** is oriented substantially in the direction of upper piezoelectric sensor **100A** while the second diagonal direction **201** is oriented substantially opposite of the direction of piezoelectric sensor **100A**. Additionally, the third diagonal direction **202** is oriented substantially opposite the direction of lower piezoelectric sensor **100C** while the fourth diagonal direction **203** is oriented substantially in the direction of piezoelectric sensor **100C**. (60) Graphs **205**, **210**, and **215** shown in FIGS. **14-16**, respectively, illustrate voltage output **161A** (graph **205**), voltage output **161B** (graph **210**), and voltage output **161C** (graph **215**) as the user **160** looks alternately in the first and second diagonal directions **200** and **201**, respectively. As shown by graphs **205**, **210**, and **25**, with upper piezoelectric sensor **100A** most closely aligned with the direction of travel of the user's eye, the voltage output **161A** of upper sensor **100A** has both the greatest positive and negative peaks while lower piezoelectric sensor **100C**, which is most offset from the direction of travel of the user's eye, has the smallest positive and negative peaks.

(61) Graphs **220**, **225**, and **230** shown in FIGS. **17-19**, respectively, illustrate voltage output **161A** (graph **220**), voltage output **161B** (graph **225**), and voltage output **161C** (graph **230**) as the user **160** looks alternately in the third and fourth diagonal directions **202** and **203**, respectively. As opposed to graphs **205**, **210**, and **215** of FIGS. **14-16**, graphs **220**, **225**, and **230** of FIGS. **17-19** illustrate that the voltage output **161C** of the lower piezoelectric sensor **100C**, which is most closely aligned to the direction of travel of the user's eye in the third and fourth directions **202** and **203**, has both the greatest positive and negative peaks. Conversely, the voltage output **161A** of the upper piezoelectric sensor **100A**, which is farthest offset from the third and fourth directions **202** and **203**, has the smallest positive and negative peaks.

(62) FIG. **20** illustrates the user **160** rotating their eye in a first (clockwise) rotational direction (indicated by arrow **231**). This rotational movement is broken down into a sequential series of steps or directional movements (indicated by arrows **232-236**) also indicated in FIG. **20**. These steps **232-236** are captured in graphs **240**, **245**, and **250** shown in FIGS. **21-23**, respectively. Particularly, graph **240** indicates voltage output **161A** of upper piezoelectric sensor **100A**; graph **245** indicates voltage output **161B** of middle piezoelectric sensor **100B**, and graph **250** indicates voltage output **161C** of lower piezoelectric sensor **100C** as the user **160** moves his eye in the first rotational direction **231**. As shown in graphs **240**, **245**, and **250**, each voltage output **161A-161C** has a series of positive and negative peaks which vary as the direction of the travel of the user's eye varies along steps **232-236**, where the degree of offset (from the direction of travel) and proximity between the sensor **100A-100C** and the then current position of the user's eye correlates with the magnitude of the given positive or negative peak.

(63) FIG. **24** illustrates the user **160** rotating their eye in a second (counterclockwise) rotational direction (indicated by arrow **251**) that is opposite the first rotational direction **231**. This rotational movement is broken down into a sequential series of steps or directional movements (indicated by arrows **252-256**) also indicated in FIG. **24**. These steps **252-256** are captured in graphs **260**, **265**, and **270** shown in FIGS. **25-27**, respectively. Particularly, graph **260** indicates voltage output **161A** of upper piezoelectric sensor **100A**; graph **265** indicates voltage output **161B** of middle piezoelectric sensor **100B**, and graph **270** indicates voltage output **161C** of lower piezoelectric sensor **100C** as the user **160** moves his eye in the second rotational direction **251**. Similar to graphs **240**, **245**, and **250** described above, as shown in graphs **260**, **265**, and **270**, each voltage output **161A-161C** has a series of positive and negative peaks which vary as the direction of the travel of the user's eye varies along steps **252-256**.

(64) Referring now to FIG. **28**, an embodiment of a flexible skin-attachable piezoelectric sensor **300**. Piezoelectric sensors **100A-100C** of eye tracking system **10** may each be configured similarly as the piezoelectric sensor **300** described below. However, the configuration of the sensors **100A-100C** of eye tracking system **10** may vary in other embodiments. The piezoelectric sensor **300** may be comfortably worn on the face of a user, such as the temple area of the user.

(65) In this exemplary embodiment, piezoelectric sensor **300** generally includes a pair of

electrically insulative layers or insulators **310** and **350**, a pair of electrically conductive layers or electrodes **320** and **340**, a piezoelectric layer or film **330**. The piezoelectric film **330** is sandwiched between the pair of electrodes **320** and **340**. Similarly, the pair of electrodes are sandwiched between the pair of insulators **310** and **350**. Piezoelectric sensor **300** also comprises an attachment pad or layer **305** configured to flexibly attach the piezoelectric sensor **300** to the face of a user such that the sensor **300** may be worn comfortably on the user's face, removed, and reattached to the user's face as needed. In some embodiments, attachment pad **305** comprises an adhesive pad meant to temporarily adhere to the user's face.

(66) The piezoelectric sensor **300** additionally includes a pair of signal conductors **352** and **354** connected to the electrodes **320** and **340**, and a wireless transmitter **360** connected to the pair of signal conductors **352** and **354**. Electrodes **352** and **354** produce a voltage output (e.g., voltage outputs **161A**, **161B**, and **161C**) in response to flexure of the piezoelectric film **330**. Wireless transmitter **360** is configured to transmit a signal corresponding to the voltage produced by signal conductors **352** and **354** to a computer system (e.g., computer system **20** shown in FIG. **1**). While in this exemplary embodiment piezoelectric sensor **300** is configured for wireless communication via wireless transmitter **360**, in other embodiments, piezoelectric sensor **300** may communicate to the computer system through a wired connection and thus may not include wireless transmitter **360**. It may also be understood that piezoelectric sensor **300** may include features in addition to those shown in FIG. **28**.

(67) The piezoelectric film **330** is flexible and does not comprise any toxic materials such as lead. Additionally, piezoelectric film **330** is highly sensitive to provide piezoelectric sensor **300** with sensitivity required to detect eye movement and position as described above with respect to eye tracking system **10**. In this exemplary embodiment, piezoelectric film **330** comprises a Gallium-Nitride-based or Gallium-Nitride-comprising material. Gallium Nitride provides a nontoxic alternative to other substances such as lead, permitting the piezoelectric sensor **300** to be safely worn on the face of the user.

(68) As will be described further herein, the piezoelectric film **330** comprises a defectless Gallium Nitride material in which defects or flaws have been intentionally removed from the Gallium Nitride material as part of forming the piezoelectric film **330** and sensor **300**. Particularly, in some embodiments, microcracks and other flaws are removed from the Gallium Nitride material as part of forming of the piezoelectric film **330**. In some embodiments, each of four edges of the Gallium Nitride material, which may contain cracks and irregular saw-tooth edges, are etched with a depth more than thickness of the Gallium Nitride film. In some embodiments, after removal of the edges, a remaining sidewall of the Gallium Nitride film defining an outer perimeter of the film has a surface roughness that is 500 micrometers (μm) or less as measured from the difference between the peaks and valleys of the sidewall of the Gallium Nitride film. Gallium Nitride is traditionally a relatively brittle material, particularly when compared with relatively flexible materials like lead used in applications in which the piezoelectric sensor must be relatively flexible. However, the lack of defects or flaws in piezoelectric film **330** provides the piezoelectric film **330** with the flexibility required to be worn on an uneven, dynamic surface such as a user's face.

(69) In addition to being more flexible than traditional Gallium Nitride materials, the Gallium Nitride film material comprising piezoelectric film **330** also has a greater sensitivity than conventional Gallium Nitride materials. For example, in some embodiments, piezoelectric film **330** has a sensitivity ranging approximately between 0.1 volts per newton (V/N) and 5 V/N. In certain embodiments, the sensitivity of piezoelectric film **330** is approximately between 0.1 V/N and 1.0 V/N. In some embodiments, the piezoelectric film **330** comprises a Gallium Nitride material having a single-crystalline structure in which there are no grain boundaries and instead the crystal lattice extends unbroken to the edges of the film **330**. For example, in some embodiments, the piezoelectric film **330** comprises a group III-nitride (III-N) thin film with a single-crystalline structure. In some embodiments, the piezoelectric film **330** comprises at least one of aluminum

nitride (AlN), gallium nitride (GaN), scandium nitride (ScN), indium nitride (InN), and combinations of these materials. In certain embodiments, piezoelectric film **330** comprises a plurality of separate and distinct layers of III-N thin films including AlN, GaN, InN, and their alloys in accordance with the formula of $\text{In}_{\text{sub.x}}\text{Al}_{\text{sub.y}}\text{Sc}_{\text{sub.z}}\text{Ga}_{\text{sub.1-x-y-z}}\text{N}$ (where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$). In certain embodiments, piezoelectric film **330** has a thickness of approximately between one nanometer (nm) and one millimeter (mm); however, it may be understood that the thickness of piezoelectric film **330** may vary in other embodiments. It may also be understood that the materials comprising piezoelectric film **330** may vary depending on the given embodiment. For example, in some embodiments, piezoelectric film **330** may not include a Gallium Nitride material.

(70) Referring to FIGS. **29-31**, an exemplary process for forming a flexible, skin-attachable piezoelectric sensor such as the piezoelectric sensor **300** is shown. In this exemplary embodiment, a single-crystalline thin film **370** (e.g., a III-N thin film) is grown epitaxially in a substrate **372** as shown in FIG. **29**. In some embodiments, during or following the growth of thin film **370** on the substrate **372**, an outer perimeter or edge **371** of the thin film **370** is detached and removed from the thin film **370**. Particularly, the entire outer edge **371** extending about the entire perimeter of the thin film **370** may be removed. Microcracks may form in the outer edge **371** of thin film **370** following the growth of film **370** on the substrate **372**. For example, in some embodiments, thin film **370** is cut into a plurality of samples following growth where the cutting process may form microcracks or other forms of damage along the outer edges **371** of the thin film **370**. If left attached to the thin film **370**, the microcracks formed in the outer edge **371** thereof may propagate through the thin film **370** as stresses are applied to the thin film **370**, limiting the durability and performance of the thin film **370** during operation of the piezoelectric sensor comprising the film **370**. In some embodiments, the outer edge **371** removed from the thin film **370** is approximately 2.0 mm or less in width. An exterior or outer sidewall **381** of the thin film **370** remaining following the removal of outer edge **371** may have a maximum or an average surface roughness that is 500 μm or less. In some embodiments, the surface roughness of outer sidewall **381** is 300 μm or less. In some embodiments, the surface roughness of outer sidewall **381** is 100 μm or less. Additionally, in certain embodiments, the thin film **370** may be analyzed (e.g., using an optical microscope) before and/or following the removal of outer edge **371** to confirm the removal of the microcracks from the thin film **370** prior to the completion of the process for forming a piezoelectric sensor from the thin film **370**.

(71) In some embodiments, substrate **372** comprises one or more of silicon (Si), sapphire, and Silicon Carbide (SiC). Following or as the thin film **370** is grown, a first or upper electrode layer **374** is deposited onto a top or upper surface **373** of the thin film **370**. The upper electrode layer **374** comprises an electrode formed from an electrically conductive material such as nickel, titanium, and gold. Following the deposition of upper electrode layer **374**, a transition layer **376** is deposited onto the electrode **374**. Transition layer **376** also comprises an electrically conductive material such as nickel or copper. In some embodiments, transition layer **376** is deposited onto upper electrode layer **374** by electroplating with a thickness ranging approximately between tens and hundreds of μm .

(72) In this exemplary embodiment, a protection layer **378** is deposited onto the transition layer **376**. Protection layer **378** comprises a noble metal such as gold. Additionally, an output of the assembly of layers is attached to a wafer **380**. In some embodiments, wafer **380** comprises a chemically stable material such as sapphire, and the wafer **380** may be attached to the protection layer **378** via an adhesive such as polymeric glue. Following the assembly of the layers (e.g., layers **370**, **374**, **376**, and **378**), the substrate **372** is removed using a microfabrication removal process such as a wet-etching process, a dry-etching process, and a laser-liftoff process.

(73) As shown particularly in FIG. **30**, following the removal of substrate **372**, a second or lower electrode layer **382** is deposited onto the now exposed bottom or lower surface **375** of the thin film

370. The lower electrode layer **382** comprises an electrode formed from an electrically conductive material such as nickel, titanium, and gold. Additionally, the wafer **380** is detached and removed from the protection layer **378**. Further, a pair of signal conductors or wires **384** and **386** are connected to the electrode layers **374** and **382**, respectively, as shown in FIG. **31**. In this arrangement, first or upper wire **384** is electrically connected to the upper electrode layer **374** while the second or lower wire **386** is electrically connected to the lower electrode layer **382**. Following the connection of wires **384** and **386** to layers **374** and **382**, a pair of insulating layers **388** and **390** are coated onto a top or upper end **389** and onto a bottom or lower end **391** of the assembly of layers to seal the electrode layers **374** and **382** and thin film **370** from the surrounding environment. In some embodiments, insulating layers **388** and **390** each comprise a Si material such as Polydimethylsiloxane (PDMS).

(74) Referring to FIG. **32**, an embodiment of a method **395** for tracking an eye of a user is shown. Initially, block **396** of method **395** includes producing signals from one or more piezoelectric sensors in response to the user moving their eye in at least one direction, wherein the one or more piezoelectric sensors are positioned on the face of the user in one or more locations each spaced from the eyelids of the user in response to the user. In some embodiments, block **396** includes producing signals from piezoelectric sensors **100A-100C** of the eye tracking system **10** shown in FIG. **1** in response to the user moving their eye in at least one direction. Block **397** of method **395** includes receiving by a computer system the signals produced by the one or more piezoelectric sensors. In some embodiments, block **397** includes receiving by the computer system **20** shown in FIG. **1** the signals produced by the piezoelectric sensors **100A-100C**. Block **398** of method **395** includes detecting by the computer system movement in the at least one direction of the eye of the user based on the signals received from the one or more piezoelectric sensors. In some embodiments, block **398** includes detecting by the computer system **20** movement in the at least one direction of the eye of the user based on the signals received from piezoelectric sensors **100A-100C**.

(75) Referring to FIG. **33**, an embodiment of a method **400** for tracking an eye of a user is shown. Initially, block **401** of method **400** includes growing a piezoelectric thin film on a first substrate, wherein the piezoelectric film comprises a Gallium material and has an outer edge extending along the perimeter of the thin film. In some embodiments, block **401** includes growing the piezoelectric film **330** of the piezoelectric sensor **300** shown in FIG. **28**. Block **402** of method **400** includes trimming at least a portion of the outer edge from the piezoelectric thin film. In some embodiments, block **402** includes trimming at least a portion of the outer edge from the piezoelectric film **330**. Block **403** of method **400** includes coupling a pair of electrically conductive electrodes to the trimmed piezoelectric thin film whereby the trimmed piezoelectric thin film is positioned between the pair of electrodes. In some embodiments, block **403** includes coupling the pair of electrodes **320** and **340** shown in FIG. **28** to the trimmed piezoelectric film **330**.

(76) Experiments were conducted regarding both flexible, skin-attachable piezoelectric sensors and systems for tracking the position and movement of a user's eye using one or more piezoelectric sensors. It may be understood that the following experiments described herein are not intended to limit the scope of this disclosure and upon the embodiments described above and shown in FIGS. **1-33**.

(77) In previous studies, strain sensors have been used to indirectly detect the movement of eyes by attaching them directly on the eyelid. The piezoelectric strain sensors based on lead zirconate titanate ($\text{Pb}[\text{Zr.sub.xTi.sub.1-x}]\text{O.sub.3}$, PZT) and zinc oxide (ZnO) were able to measure the strain change on the eyelids upon their moving; however, they showed limitations as a wearable and reliable sensor. Particularly, PZT contains a significant amount of harmful lead, which causes poisoning in the human body. Wearable and implantable sensors should generally be toxic-element-free. Additionally, ZnO sensor showed low sensitivity and resolution, which requires additional signal processing of amplification and noise reduction. The benefit of non-toxicity of ZnO cannot

compensate for low sensing performance, the most fundamental function of the sensors. Most importantly, all the previous sensors were attached directly on the eyelid where the most pronounced output signals from the surface strain change can be obtained at the expense of comfort and safety of the user. For at least this reason, the previous sensors are not truly noninvasive for safe and comfortable wearing.

(78) Alternative piezoelectric sensing elements have been sought for in healthcare monitoring systems to replace safety-hazard-ridden PZT sensors while achieving high electromechanical coupling factors comparable to PZT. The current study investigates single-crystalline group III-N materials, especially gallium nitride (GaN) thin films, which have the potential for excellent piezoelectric sensing and energy-harvesting with many advantageous electrical, mechanical, and chemical properties. GaN thin films, naturally formed permanent electric dipoles are already aligned in one direction, hence, electrical poling process is not required to obtain piezoelectric property, which is different from PZT. Additionally, GaN is chemically and thermally stable and generally safe on human skin and in the body. Further, GaN is stable and does not react with skin and body fluid. The biocompatibility of GaN makes it a prime candidate to overcome the critical limitation of PZT in epidemic and implantable sensors and electronics. Moreover, GaN thin films on flexible substrates showed excellent mechanical bendability and durability without degradation in materials and output characteristics of the devices, which provides an important benefit for reliable and continuous monitoring of the sensor. This characteristic is somewhat surprising considering the brittle nature of its bulk material; however, the durability and bendability of the film have been experimentally confirmed as part of this study. Most importantly, GaN has a relatively high electromechanical coupling coefficient due to its low dielectric constant. Single-crystalline GaN thin films are indicated by this study as exhibiting high performance in pressure sensing (output value, sensitivity, response time, and stability).

(79) In the present study, the operating principles and characteristics of the sensor for eye blinking and eyeball motions were investigated. The output signals generated by the sensor due to eye motion were numerically estimated, experimentally measured, and analyzed at various positions (upper eyelid, lower eyelid, and temple). As a result, it was demonstrated for the first time that an eye movement sensor attached on a temple area of the face is capable of accurately measuring various movements of eyes for noninvasive and reliable sensing in many extended applications.

(80) Sensor Design and Fabrication

(81) Referring now to FIG. 34, a process 408 is shown for fabricating flexible piezoelectric eye-movement sensors (F-PEMS). An edge area within ~1 mm from the edge of the III-N thin films comprising the F-PEMS were removed. These edge areas were damaged during the dicing process. Microcracks existing in the edge area could be propagated by mechanical stresses during the handling, attachment, and operation of the sensor, potentially resulting in the degradation of sensor performance. Enhanced reliability and durability of the sensor could be achieved by this process step. Optical microscope images of as-diced and edge-removed samples of GaN on Si confirmed that microcracks in the edges within several hundred micrometers no longer existed after the edge removal. Then, an electrode comprising a stack of nickel, gold, copper, and gold (Ni/Au/Cu/Au) metal layers was deposited on top of the III-N thin-film surface, which became a bottom electrode after the flip of the sample for further fabrication. A sapphire substrate was bonded to the electrode of the sample and the Si substrate was etched. Then, the sapphire was detached from the sample. After removing the Si and sapphire substrates, the sample was flipped and a top electrode with Ni/Au was deposited to complete the fabrication of the sensor element. Finally, the device was sealed by polydimethylsiloxane (PDMS) with wires attached to the electrodes. The top and bottom electrodes became anode and cathode, respectively.

(82) Referring now to FIGS. 35-37, a fabricated F-PEMS 410 produced as part of this study is shown in FIG. 35. F-PEMS 410 having a flexible single-crystalline III-N film 412 sandwiched between the electrodes 414 and 416 encapsulated in PDMS 418. The F-PEMS 410 is conformal for

the attachment of any skin surface and small enough for minimally invasive sensing. Furthermore, the output signal of the sensor-PEMS **410** was higher than that of a rigid sensor with uniform deformation. F-PEMS **410** was tested under flat and bending test conditions using a scanning-electron microscopy (SEM). The F-PEMS **410** surfaces were crack-free without surface features regardless of bending conditions, confirming that no bulk defect was generated even with a significant degree of bending.

(83) Characteristics of Piezoelectric Sensing Element

(84) The crystalline orientation and quality of the piezoelectric sensing element were investigated by high-resolution X-ray diffraction (HR-XRD). The III-N film on the Si substrate was epitaxially grown to produce the single-crystalline wurtzite structure with a surface of (0001) plane. A III-N film's out-of-plane growth direction is aligned in a [0001] direction of wurtzite structure, which is confirmed by the strong presence of (0001) peaks, such as (0002) and (0004). Therefore, a d33 piezoelectric structure was constructed, since the sensing element is sandwiched between two electrodes. Very sharp (0002) peaks of the III-N layers for both rigid and flexible substrates indicated the high crystalline quality of the sensing element before and after the fabrication of the device. Moreover, no other peak of wurtzite GaN, Al.sub.xGa.sub.1-xN, and AlN was observed in the 2 θ - ω scan of XRD. Six sharp peaks were located at every 60° in the rotational scans (ϕ scan) around GaN {1012} planes, i.e., six-fold rotational symmetry of the hexagonal-lattice structure (wurtzite), indicating that all the grains perfectly aligned in both in-plane and out-of-plane directions. Such XRD results confirm the single-crystallinity of the III-N film. After the transfer process of the III-N film, the peaks were maintained, which indicated no structural changes.

(85) Referring to FIGS. **36** and **37**, graphs **420** and **425** are shown. Particularly, graph **420** illustrates rocking curves for a single-crystalline III-N thin film on a rigid (Si) substrate (curve **421**) and a flexible (Cu/PDMS) substrate (curve **422**) for the GaN (0002) plane. Graph **425** similarly illustrates rocking curves **421** and **422** for the III-N thin film on the rigid substrate (curve **426**) and flexible substrate (curve **427**) for the (1012) plane. For the single-crystalline thin films, the density of crystalline line defects were estimated from the rocking curves (ω scans) of symmetric and asymmetric planes, as shown in graphs **420** and **425** of FIGS. **36** and **37**, respectively. The full-width at half-maximum values of the GaN (0002) and GaN (1012) peaks on Si (curves **421** and **426** of graphs **420** and **425**, respectively) were 336 arc-seconds (arc-sec.) and 828 arc-sec., respectively. The density of threading dislocations of the single-crystalline GaN film was calculated to be approximately $3.2 \times 10^9 \text{ cm}^{-2}$. Such a low density of line defects in single-crystalline material where no grain boundary exists is important to reduce the leakage between the electrodes and to enhance the sensitivity of the sensor. The peaks in the rocking curves of the flexible substrate (curves **422** and **427** of graphs **420** and **425**, respectively) were broader than those of the Si substrate. This phenomenon was attributed to the wavy or curved nature of the sample rather than the degradation of crystalline quality itself. The intensities of GaN peaks on a flexible substrate, i.e., after the fabrication were lower than those on Si substrate, possibly due to the buried location of GaN under a top electrode.

(86) Sensing Mechanism of Eye Movements

(87) Referring to FIGS. **38** and **39**, as part of this study, a piezoelectric sensor **430** (i.e., a F-PEMS) was attached on a curved part of the skin near the eyes of a test subject. After the attachment, the sensor **430** was initially slightly bent depending on the contour of the skin of a face (shown at the middle of FIG. **39**). When the bending degree of the sensor **430** was changed by the movement of the skin or muscles underneath, the change in piezoelectric polarization of the III-N film of the sensor **430** occurred. Because the bottom electrode of sensor **430** was significantly thicker than the III-N film, a neutral plane (shown in FIG. **38**) of the sensor **430** was located in the Cu layer of the electrode (shown in FIG. **38**). This configuration induced only one type of piezoelectric polarization change (positive or negative) depending on the bending mode.

(88) When the sensor was further bent (shown in FIG. **39**), in-plane tensile strain was applied and

an upward piezoelectric polarization was accordingly induced. As a result, positive and negative charges were induced on the top (N-face) and bottom (Ga-face) surfaces, respectively. The electrical current flowed from top to bottom electrodes through the external wires because of the free electron movement to cancel out the piezoelectric charges on the surfaces of GaN (shown at the top of FIG. 39). Thus, the positive voltage output was measured. When the sensor was flattened, an in-plane compressive strain was applied and the negative voltage was measured (shown at the bottom of FIG. 39). Further bending and flattening resulted in opposite signs in the output voltage of the sensor 430. The change in skin contour by the eye movement is subtle; therefore, the piezoelectric sensor should be very sensitive to generate a substantial magnitude of the signal output voltage. Mathematical modeling software (COMSOL Multiphysics® and MATLAB®) was used to theoretically predict the output. Several parameters were applied including elastic modulus, Poisson ratio, unit weight, the thickness of film, sensor dimension (x and y), and applied load. Referring to FIG. 40, a graph 440 is shown illustrating the numerically simulated deflection and corresponding output voltages 441 of the piezoelectric sensor 430, confirming that the sensor 430 was sensitive enough to measure the subtle contour change.

(89) As described above, accurate and reliable sensing of eye movement in a noninvasive and comfortable way is important, therefore, the sensing on a temple area over upper and lower eyelid areas was preferred during the study. While the strain change on the skin of upper and lower eyelids is relatively straightforward, the skin contour change of the temple area is associated with muscle movement. Referring to FIG. 41, the bending status of the piezoelectric sensor 430 for contraction and release modes is shown.

(90) The piezoelectric sensor 430 can be further bent or flattened from the initial condition by the eye blinking or eyeball motion by contraction and release force from muscle movement. Therefore, the modes of eye movements generate different output voltages, i.e., opposite signs and varied magnitude. For the eye blinking, the main protractor of the eyelid, which serves to close the eye, is the orbicularis oculi and it is connected to the temple via ligament. The ligament underneath of the temple is tightened or loosened for opening or closing of the eyelid during the eye-blinking process, respectively. For the eyeball movements, different muscles are involved. Among these muscles, lateral rectus and medial rectus are principal parts for side-glance of the eyeball, which can be displayed with simulated results. The muscles and ligament underneath the temple area are contracted when the eyeball moves toward the temple side (right) and are expanded when the eyeball moves away from the temple side (left).

(91) Eye-Blink Sensing

(92) Referring to FIGS. 42-51, sensing characteristics for eye-blink movements are shown indicating that the F-PEMS (e.g., F-PEMS 410, piezoelectric sensor 430) yields substantial output voltage changes with various blink behaviors. Particularly, FIG. 42 shows three locations around the eye including images of a test subject 450 wearing the F-PEMS on the upper eyelid 451, lower eyelid 452, and temple 453 using a biocompatible tape. This adhesive material was strong enough to bond on human skin with high flexibility. It should be noted that the size of the F-PEMS (1×1 centimeters squared (cm.sup.2)) was determined by the dicing size of the GaN/Si sample. The F-PEMS can be made significantly smaller without affecting the output voltage signal. The voltage output from sensors with different sizes including 0.25 cm.sup.2 (0.5×0.5 cm.sup.2) and 4 cm.sup.2 (2×2 cm.sup.2) was confirmed experimentally as being similar to those of the F-PEMS used in this study.

(93) FIGS. 45-47 illustrate graphs 455, 460, and 465, respectively, of output voltages 456, 461, and 466 respectively, produced by the F-PEMS when positioned on the upper eyelid 451. FIGS. 46-48 illustrate graphs 470, 475, and 480, respectively, of output voltages 471, 476, and 4481 respectively, produced by the F-PEMS when positioned on the lower eyelid 452. FIGS. 49-51 illustrate graphs 485, 490, and 495, respectively, of output voltages 486, 491, and 496, respectively, produced by the F-PEMS when positioned on the temple 453. Additionally, graphs 455, 470, and

485 illustrate overall sensing characteristics; graphs **460**, **475**, and **490** illustrate a single eyelid opening/eyelid closing event; and graphs **465**, **480**, and **495** illustrate eyelid motion including overlong closure and overlong gazing instead of eye blinking.

(94) The overall average values of the output voltages were 97.85 millivolts (mV) and -96.02 mV (illustrated by graphs **455** and **460**) for closing and opening when the F-PEMS was positioned on the upper eyelid **451**; 78.31 mV and -79.42 mV for closing and opening when the F-PEMS was positioned on the lower eyelid **452** (illustrated by graphs **470** and **475**); and 32.3 mV and -30.3 mV for closing and opening when the F-PEMS was positioned on the temple **453** (illustrated by graphs **485** and **490**). The absolute values in the change were highest for the upper eyelid **451** and lowest for the temple **453**, which is related to the difference in the crooked level of the face parts. It is not surprising that the previous studies only show measurements from the eyelid area, which causes discomfort while wearing for the measurements. In this study, Measurement from the temple **453** was demonstrated in this study. Signal-to-noise ratios (SNR, S/R) were 49.26 decibels (dB), 47.88 dB, and 40.13 dB for the measurements on the upper eyelid **451**, lower eyelid **452**, and temple **453**, respectively. Not intending to be bound by any particular theory, the SNR was determined in accordance with the following Equation where V_{signal} represents the average value of output voltages and V_{noise} represents the averaged background signals:

$$(95) \text{ SNR}_{\text{dB}} = 20\log\left(\frac{V_{\text{signal}}}{V_{\text{noise}}}\right) \quad (1)$$

(96) All the SNR values were greater than 40 dB, suggesting that the voltage values from the F-PEMS on the temple **453** were sufficient to communicate in wireless data.

(97) Graphs **460**, **475**, and **490** generated the positive and negative voltage outputs for the eyelid closing and opening, respectively, enabling to distinguish four stages of a normal eye blink motion, i.e., closing, closure, opening, and gazing. The eye closing corresponds to the output positively rising from zero to the peak. The output drops to and remains at zero until the further movement of the eyelid, corresponding to eye closure. The changes in voltage output **461**, **476**, and **491** were in the opposite direction with increasing magnitude during the eye-opening, i.e., moving from zero to the valley. The voltage outputs **461**, **476**, and **491** changed to and remained at zero during the gazing stage. The rapid eye blinking was also measured with higher frequency (0.24 Hertz (Hz). f.wdarw.1.67 Hz). The flickering rate was influenced by short-term changes in the brain, such as tiredness and/or sleepiness. The voltage values were very similar to the outputs of blinking at a normal rate. These results indicated that the F-PEMS can be utilized to detect various eyelid movements for the evaluation of normal blinking, dry eyes, and drowsiness.

(98) Long-term stability is one of the key parameters for the operation of sensors. The F-PEMS was tested on the skin of a human test subject by continuous monitoring of eye blinking of the subject for 10 minutes (min) followed by regular measurement every 30 min for approximately 10 hours (h). During the test, the subject wore the sensor during their daily routine without discomfort thanks to the conformal, compact, and noninvasive nature of the F-PEMS. No degradation of the voltage output was observed for both the continuous monitoring and long-term monitoring. No performance issue was expected for long-term continuous monitoring. The F-PEMS can be reused by the subject in the following days. The stability and durability of the F-PEMS were thus confirmed.

(99) Fatigue and Drowsiness Detection

(100) The degree of fatigue and drowsiness of a subject can be estimated by blink frequency, blink duration (delays of eye reopening), and PERCLOS (percentage of eye closure). A lid closure that lasts more than about 500 milliseconds (ms) and covers the pupil for that time is usually defined as a microsleep. To simulate the state of the microsleep, overlong closure was investigated using the same positions. The overlong closure is of great interest in assessing workers, warfighters, and driving safety to detect blind working, blind walking, and drowsy driving, respectively. Graphs **465**, **480**, and **495** illustrate that the F-PEMS was capable of detecting microsleep in the test subject

in real-time. As shown in graphs **465**, **480**, and **495** sections of overlong closure and gazing are distinguished by the voltages upon the initiation of eyelid movement. For example, the gap between the positive and negative potentials was “overlong closure,” whereas the interval between the negative and positive outputs was “gazing.” Unlike the computer-vision-based ETS, the detection does not require additional calibration. The F-PEMS can directly provide the duration time for the overlong closure from the output change. Also, the response time of the F-PEMS was brief enough to distinguish the four stages of eye blinking.

(101) Tiredness in an individual can be recognized through changes in the individual's eyes, especially the blinking of eyes. Also, when people are exhausted, their eyes typically close unconsciously while walking and driving. To investigate the fatigued eye, we simulated the status of dozing off was simulated whereby the test subject closed their eyes halfway (here, almost half and 75% closed) and blinked their eyes alternating between normal blinks and half-blinks.

(102) Referring to FIGS. **52** and **53**, graphs **500** and **505** are shown illustrating voltage outputs **501** and **506**, respectively, generated by the F-PEMS when positioned on the test subject's eye with the subject half-blinking in order to mimic a fatigued eye. Additionally, graph **500** illustrates the response of the F-PEMS when the subject closes 50% of their eye when half-blinking while graph **505** illustrates the response of the F-PEMS when the subject closes 75% of their eye when half-blinking. The voltage outputs **501** and **506** were relatively small value approximately 50% for output **501** and 72% for output **506**), which means that we could discern the condition of the eye from voltage outputs **501** and **506**. The results from other parts (upper eyelid and lower eyelid) with the half-closed condition showed a similar tendency to the voltage outputs from the temple area.

(103) Eyeball-Movement Sensing

(104) In addition to the sensing of eye blink for the detection of eye fatigue and drowsiness, the F-PEMS can measure the movement of the eyeball, which is related to sleeping patterns, the additional diagnosis of brain-related diseases, and eye tracking in AR/VR systems. Referring to FIGS. **54** and **55**, graphs **510** and **515** are shown illustrating voltage outputs **511** and **516**, respectively, generated by the F-PEMS when positioned on the test subject's eye with the subject looking repeatedly from a center (C) position, a right (R) position, and a left (L) position. Additionally, graph **510** illustrates the response of the F-PEMS when the subject moves their eye slowly while graph **515** illustrates the response of the F-PEMS when the subject moves their eye rapidly.

(105) When the eyeball moved to the right, i.e., moving toward the sensor side, positive values were measured. In contrast, when the eyeball moved in the opposite direction, i.e., moving away from the sensor side, negative values were obtained. This output is different from the signal of the EOG that measures the potential difference between cornea and retina, which is influenced by numerous external conditions, e.g., light intensity, number of electrodes, skin status, and noise from contact resistance. As the contraction of the lateral rectus pulled the test subject's eye away from the nose (right seeing), the sensor was further bent. Conversely, as the contraction of the medial rectus pulled the test subject's eye toward the nose (left seeing), the sensor was flattened. The eyeball moving to the right and the left sides corresponded to contraction and release of the F-PEMS, respectively.

(106) Additionally, when the eyeball moved toward the sensor side, further bending of the sensor occurred, hence positive voltage outputs (voltage outputs **511** and **516**) were generated. When the eyeball moved away from the sensor side, the negative voltage output was generated. The output characteristics were measured in both cases of slow (voltage output **511**) (≈ 0.5 Hz) and rapid (voltage output **516**) (≈ 1.9 Hz) eye movement, showing similar output voltage values regardless of the speed of the motion. In graph **515**, the initiation of the lateral eye movement can be detected with a fast response time. The output from different sensor positions (upper eyelid and lower eyelid) showed a similar trend only with higher output voltage values, attributed to relatively larger

deformation of the F-PEMS. The overall values of voltage output were different for eyelid movements (≈ 30 mV) and eyeball movements (≈ 10 mV). Therefore, each movement was distinguishable for the sensing of combined movements of eye blinking and eyeball moving. Moreover, further detailed movement of eyeballs (vertical and oblique directions and rotational movements) and eyelid (left and right separately) could be detected if arrayed (e.g., 1×3) sensors were attached on both sides of the temples.

CONCLUSIONS OF THE STUDY

(107) A flexible piezoelectric sensor or F-PEMS was designed, fabricated, and tested to develop a personal safety, behavior, and healthcare monitoring sensing network. A stable and safe piezoelectric single-crystalline III-N thin film was epitaxially grown on a Si (111) substrate and then transferred to a flexible PDMS substrate followed by the electrode formation. The fabricated F-PEMS was tested on several different parts around the eyes, including the upper eyelid, lower eyelid, and temple area. The F-PEMS was estimated to generate substantial output voltage signals enough to detect small deflections of muscles and skins caused by eyelid and eyeball movements based on numerical simulation and demonstrated to measure normal eye blinking with signal-to-noise ratios of higher than 40 dB, confirming high sensitivity and usability for wireless communication systems. The F-PEMS was especially capable to detect various movements of eyelid and eyeballs from the most-comfortably-wearable temple area. This noninvasive, nontoxic, and easily-wearable F-PEMS was demonstrated to detect eye blink frequency, blink duration, and percent of eye closure to function as an objective indicator of eye strain, fatigue, and drowsiness. The F-PEMS was also demonstrated to measure lateral movements of eyeballs, which are distinguished from the eye blink, for the potential use of medical diagnosis indicators of autism, ADHD, stroke, Alzheimer and Parkinson's disease as well as in industrial applications such as VR and AR in remote control of robots.

(108) While embodiments of the disclosure have been shown and described, modifications thereof can be made by one skilled in the art without departing from the scope or teachings herein. The embodiments described herein are exemplary only and are not limiting. Many variations and modifications of the systems, apparatus, and processes described herein are possible and are within the scope of the disclosure. For example, the relative dimensions of various parts, the materials from which the various parts are made, and other parameters can be varied. Accordingly, the scope of protection is not limited to the embodiments described herein, but is only limited by the claims that follow, the scope of which shall include all equivalents of the subject matter of the claims. Unless expressly stated otherwise, the steps in a method claim may be performed in any order. The recitation of identifiers such as (a), (b), (c) or (1), (2), (3) before steps in a method claim are not intended to and do not specify a particular order to the steps, but rather are used to simplify subsequent reference to such steps.

Claims

1. A system for tracking an eye of a user, the system comprising: one or more piezoelectric sensors positionable on the face of the user and into contact with the user's skin; and an eye tracking computer system in signal communication with the one or more piezoelectric sensors so as to receive signals from the one or more piezoelectric sensors; wherein the computer system is configured to detect movement in at least one direction of the eye of the user based on the signals received by the computer system from the one or more piezoelectric sensors in response to flexure of the one or more piezoelectric sensors when positioned on the face of the user in one or more locations each spaced from the eyelids of the user.
2. The eye tracking system of claim 1, wherein the computer system is configured to detect the movement in the at least one direction of the eye of the user when the one or more piezoelectric sensors are positioned on the temple of the user.

3. The eye tracking system of claim 1, wherein: the one or more piezoelectric sensors comprises a plurality of the piezoelectric sensors positionable on the face of the user; and the computer system is configured to detect movement of the eye of the user in both a lateral direction and a vertical direction orthogonal to the lateral direction when the plurality of the piezoelectric sensors are positioned on the face of the user in a plurality of locations each spaced from the eyelids of the user.
4. The eye tracking system of claim 3, wherein the computer system is configured to compare the signals produced by the plurality of the piezoelectric sensors to detect the movement of the eye of the user in both the lateral direction and the vertical direction when the plurality of the piezoelectric sensors are positioned on the face of the user in the plurality of locations each spaced from the eyelids of the user.
5. The eye tracking system of claim 3, wherein the computer system is configured to detect movement of the eye of the user in a diagonal direction that is at a non-zero angle to both the lateral direction and the vertical direction when the plurality of the piezoelectric sensors are positioned on the face of the user in the plurality of locations each spaced from the eyelids of the user.
6. The eye tracking system of claim 1, wherein the computer system is configured to detect movement of the eye of the user in a rotational direction when the plurality of the piezoelectric sensors are positioned on the face of the user in the plurality of locations each spaced from the eyelids of the user.
7. The eye tracking system of claim 1, wherein the signals produced by the one or more piezoelectric sensors are contingent upon the deflection of the facial skin of the user upon which the one or more piezoelectric sensors are positioned.
8. The eye tracking system of claim 1, wherein each of the one or more piezoelectric sensors comprises a pair of electrodes and a piezoelectric film positioned between the pair of electrodes.
9. The eye tracking system of claim 8, wherein the piezoelectric film comprises at least one of aluminum nitride, gallium nitride, and indium nitride.
10. The eye tracking system of claim 8, wherein each of the one or more piezoelectric sensors comprises an outer insulating layer sealing the piezoelectric film and the pair of electrodes from the external environment.
11. The eye tracking system of claim 8, wherein each of the one or more piezoelectric sensors comprises an adhesive pad for removably attaching the one or more piezoelectric sensors to the face of the user.
12. The eye tracking system of claim 1, wherein: the computer system is configured to generate an image based on the detection of the movement of the eye of the user in the at least one direction; and the computer system comprises a visual display configured to indicate the image to the user.
13. A method for tracking an eye of a user, the method comprising: (a) producing signals from one or more piezoelectric sensors in response to the user moving their eye in at least one direction, wherein the one or more piezoelectric sensors are positioned on the face of the user and in contact with the user's skin in one or more locations each spaced from the eyelids of the user in response to the user; (b) receiving by a computer system the signals produced by the one or more piezoelectric sensors; and (c) detecting by the computer system movement in the at least one direction of the eye of the user based on the signals received from the one or more piezoelectric sensors in response to flexure of the one or more piezoelectric sensors positions on the face of the user.
14. The method of claim 13, wherein: (a) comprises producing signals from a plurality of the piezoelectric sensors in response to the user moving their eye in at least one direction, wherein the one or more piezoelectric sensors are positioned on the face of the user in one or more locations each spaced from the eyelids of the user in response to the user; (b) comprises receiving by the computer system the signals produced by the plurality of the piezoelectric sensors; (c) comprises detecting by the computer system movement of the eye of the user in both a lateral direction and a vertical direction orthogonal to the lateral direction; and optionally (c) comprises detecting by the

computer system movement of the eye of the user in a diagonal direction that is at a non-zero angle to both the lateral direction and the vertical direction.

15. The method of claim 13, wherein: (a) comprises producing signals from a plurality of the piezoelectric sensors in response to the user moving their eye in at least one direction, wherein the one or more piezoelectric sensors are positioned on the face of the user in one or more locations each spaced from the eyelids of the user in response to the user; (b) comprises receiving by the computer system the signals produced by the plurality of the piezoelectric sensors; and (c) comprises detecting by the computer system movement of the eye of the user in a rotational direction.
